Problem

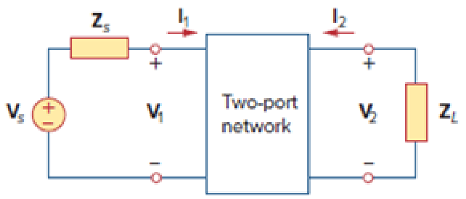
For the two-port network shown in Fig. 19.75, show that at the output terminals,

$$\mathbf{Z}_{Th} = \mathbf{z}_{22} - \frac{\mathbf{z}_{12}\mathbf{z}_{21}}{\mathbf{z}_{11} + \mathbf{Z}_s}$$

and

$$\mathbf{V}_{Th} = \frac{\mathbf{z}_{21}}{\mathbf{z}_{11} + \mathbf{Z}_s}\mathbf{V}_s$$

Figure 19.75

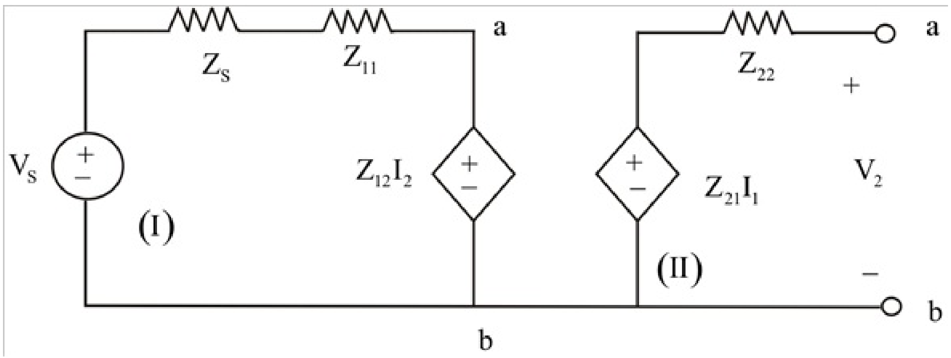


Step-by-step solution

Step 1 of 3

For the two port $\frac{N}{W}$ we have:-

$$V_1 = Z_{11}I_1 + Z_{12}I_2$$
$$V_2 = Z_{21}I_1 + Z_{22}I_2$$



Step 2 of 3

In loop I

$$I_1 = \frac{Vs - Z_{12}I_2}{Z_s + Z_{11}}$$

In loop II

$$V_2 = (Z_{21}I_1 + Z_{22}I_2)$$

Substituting from above

$$V_2 = \frac{Z_{21}}{Z_s + Z_{11}}(V_s - Z_{12}I_2)$$

Step 3 of 3

Dividing above equation by I_2

$$\frac{V_2}{I_2} = \frac{Z_{21}}{Z_s + Z_{11}}\left(\frac{V_s}{I_2} - Z_{12}\right) + Z_{22}$$

For Z_{th} $V_s = 0$, and $\frac{V_2}{I_2} = Z_{th}$ ______(1)

$$Z_{th} = -\left(\frac{Z_{21}Z_{12}}{Z_s + Z_{11}} - Z_{22}\right)$$

$$V_{th} = V_2 = {}_{21}I_1 + Z_{22}I_2$$

Since AB is open circuit $I_2 = 0$,

$$V_{th} = \frac{Z_{21}}{Z_{11} + Z_s} V_s$$

Hence for the above circuit

$$Z_{th} = Z_{22} - \frac{Z_{12}Z_{21}}{Z_{11} + Z_s} V_s$$